

HIGH ACRYLONITRILE NBR: PATENT LANDSCAPE ON SYNTHESIS, MOLECULAR-WEIGHT CONTROL, AND POST-POLYMER PROCESSING

1. ABSTRACT

This report synthesizes a patent set relevant to high acrylonitrile nitrile butadiene rubber (NBR), with emphasis on how acrylonitrile-rich compositions are made, isolated, modified, and processed. The evidence spans classic emulsion-polymerized NBR, solution-polymerized nitrile rubbers, and downstream handling routes that preserve or improve properties needed for oil-resistant elastomers. The core target is NBR containing a relatively high proportion of acrylonitrile units, typically disclosed in the context of improved oil resistance, heat resistance, processability, or compatibility with subsequent hydrogenation or compounding. Across the patents, the dominant synthesis route remains free-radical polymerization of acrylonitrile with butadiene, most often in emulsion, but solution polymerization is also disclosed as a route to better control molecular weight and architecture.

Several recurring process themes emerge. Older patents emphasize conventional emulsion or mass/mass-suspension polymerization with molecular-weight regulators such as alkylthiols or dodecyl mercaptans, and initiators selected for thermal windows compatible with high conversion and manageable viscosity. Later patents focus on more controlled architectures, including in-situ emulsifier formation, seed-latex strategies, and continuous polymerization to narrow molecular-weight distribution. One patent discloses solution polymerization of nitrile rubbers in organic solvent with special modifiers, explicitly addressing the long-standing industrial challenge of making NBR directly in solution. Another patent describes a continuous mass process for ABS-type polyblends that includes acrylonitrile and butadiene monomers, showing the broader utility of acrylonitrile/butadiene feedstocks in grafted systems. A more recent patent set shifts from synthesis to processing: one patent improves hydrogenated nitrile rubber isolation by a two-step coagulation sequence using halogen-free and divalent metal salts, while another uses seed-latex emulsion polymerization with at least 22 wt-% nitrile monomer for dip-molding latexes. Overall, the landscape indicates that high-ACN NBR is most robustly produced by emulsion polymerization with mercaptan-based molecular-weight control, but newer approaches increasingly prioritize controlled polymerization, cleaner coagulation, and process integration to support downstream hydrogenation and molding.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US7592409B2
- Patent Title: Styrene acrylonitrile isoprene triblock copolymer
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2009
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 2

- Patent Number: US11136418B2
- Patent Title: Method for manufacturing hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 3

- Patent Number: EP2478021B1
- Patent Title: Nitrile rubbers and production of same in organic solvents
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 4

- Patent Number: US5807941A
- Patent Title: Unsaturated nitrile-conjugated diene copolymer process for producing same and ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 1998
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 5

- Patent Number: US12359010B2
- Patent Title: Polymer latex for dip-molding applications
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2025
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 6

- Patent Number: US4239863A
- Patent Title: Process for the polymerization of acrylonitrile-butadiene-styrene resins
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 1980
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 7

- Patent Number: US3903200A
- Patent Title: Continuous mass polymerization process for ABS polymeric polyblends
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 1975
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 8

- Patent Number: US11130832B2
- Patent Title: Modified conjugated diene-based polymer and rubber composition including the ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 9

- Patent Number: US10336853B2

- Patent Title: Polymer, process and composition
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 10

- Patent Number: US10731020B2
- Patent Title: Rubber composition for vulcanization molding, process for manufacturing the ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2020
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US5807941A and EP2478021B1 are the most directly relevant synthesis disclosures for NBR: both center on acrylonitrile/butadiene copolymerization, but US5807941A emphasizes emulsion polymerization with an alkylthiol molecular-weight modifier added in two stages, whereas EP2478021B1 emphasizes solution polymerization in organic solvent with special modifier compounds.
- US5807941A discloses 10 to 60 wt-% unsaturated nitrile units, preferably 20 to 50 wt-%; EP2478021B1 does not provide a specific acrylonitrile range in the extracted text, but it explicitly covers at least one conjugated diene and at least one α,β -unsaturated nitrile in solution.
- US5807941A uses a free-radical initiator and an alkylthiol compound with 12 to 16 carbon atoms and at least three tertiary carbon atoms; 10 to 95% of the modifier is charged before polymerization and the remainder is added when conversion reaches 20 to 70%. EP2478021B1 uses free-radical polymerization in organic solvent with a modifier, and optionally hydrogenation, but the extracted text does not specify the modifier identity or feed strategy.
- US5807941A targets Mooney viscosity of 15 to 150 and a low-molecular-weight fraction of 3 to 20 wt-% with Mn not larger than 35,000; EP2478021B1 focuses on solution-processable nitrile rubbers and on structural elements originating from the regulator compounds, indicating a stronger emphasis on architecture control than on a specific Mooney window in the extracted passages.
- US4239863A and US3903200A are not NBR-only patents, but both include acrylonitrile/butadiene-containing systems. US4239863A is a mass or mass-suspension ABS process using tertiary-butyl peroxy esters with 10-hour half-lives between about 50 and 95 °C, initial plug-flow polymerization up to about 10 wt-% solids, then agitation from about 5 to 10% conversion to about 40% conversion, and overall 50 to 90 wt-% monomer conversion. US3903200A is a continuous mass process for ABS-type polyblends using 0.1 to 15 wt-% soluble diene rubber, 10 to 40% conversion in the first stage, then 25 to 80% conversion in the second stage, and devolatilization to a bimodal rubber-particle morphology.

- US11136418B2 and US12359010B2 are downstream/process patents rather than primary NBR synthesis disclosures. US11136418B2 starts from nitrile rubber latex made by emulsion polymerization, then uses a first coagulation with a halogen atom-free metal salt, solution hydrogenation, and a second coagulation with a divalent metal salt; it specifies hydrogenated nitrile rubber with iodine value 120 or less and nitrile units of 10 to 60 wt-% in the precursor family. US12359010B2 uses free-radical emulsion polymerization of at least one conjugated diene and at least one ethylenically unsaturated nitrile compound in the presence of seed latex particles, with the nitrile monomer at least 22 wt-% based on total ethylenically unsaturated monomers.
- US11130832B2 and US10336853B2 are broader polymer-process patents that inform the landscape but are not direct NBR synthesis disclosures. US11130832B2 emphasizes continuous polymerization, narrow molecular-weight distribution, and terminal functionalization in conjugated diene polymers. US10336853B2 focuses on low-Mn, high-Tg solution polymers based on itaconate monomers, with lower initiator, lower CTA, and lower temperature needs; it is not an NBR patent but illustrates the broader trend toward solution-process control and reduced reliance on styrene-like monomers.
- US7592409B2 is a controlled emulsion polymerization patent that explicitly includes acrylonitrile/methacrylonitrile and conjugated diene blocks in block copolymer architectures, but it is not an NBR process patent. Its relevance is methodological: in-situ emulsifier formation and controlled radical polymerization concepts are adjacent to modern efforts to control nitrile-containing elastomer architectures.
- US10731020B2 is a rubber compounding and vulcanization-molding patent that lists nitrile rubber and acrylonitrile-butadiene rubber among possible base rubbers, but it does not disclose NBR synthesis conditions. It is relevant only as a downstream processing reference.
- Across the primary patents, the recurring historical approach is emulsion polymerization with mercaptan chain transfer and conventional coagulation; newer approaches add controlled polymerization, solution synthesis, seed-latex control, and cleaner coagulation/hydrogenation workflows.

4. CONCLUSION

The patent evidence supports high acrylonitrile NBR as a family most reliably made by free-radical copolymerization of acrylonitrile with butadiene, with emulsion polymerization remaining the industrial baseline and mercaptan-based molecular-weight control the most explicit recurring tactic. For higher-ACN grades, the most defensible synthesis window from the extracted record is an acrylonitrile content in the broad range of about 10 to 60 wt-% of nitrile units, with a preferred practical band around 20 to 50 wt-% where disclosed. For process control, the strongest evidence favors staged addition of chain-transfer agent, careful initiator selection, and conversion management to keep viscosity and molecular-weight distribution within workable limits. Where solution processing is desired, the patents indicate that organic-solvent free-radical polymerization with special modifiers can be used, but the evidence also shows that this route is less established than emulsion polymerization. For downstream performance, the best-supported workflow is

emulsion polymerization followed by coagulation, isolation, and, if needed, hydrogenation using a two-step coagulation strategy with halogen-free and divalent metal salts. Overall, the recommended evidence-based synthesis strategy for high acrylonitrile NBR is emulsion polymerization with controlled mercaptan chain transfer, moderate-to-high acrylonitrile feed, and post-polymer isolation conditions designed to preserve molecular weight and enable clean conversion to HNBR when required.

5. REFERENCES

- US7592409B2 | Styrene acrylonitrile isoprene triblock copolymer | Not disclosed | US | 2009 | <https://patents.google.com/patent/US7592409B2/en>
- US11136418B2 | Method for manufacturing hydrogenated nitrile rubber | Not disclosed | US | 2021 | <https://patents.google.com/patent/US11136418B2/en>
- EP2478021B1 | Nitrile rubbers and production of same in organic solvents | Not disclosed | EP | 2015 | <https://patents.google.com/patent/EP2478021B1/en>
- US5807941A | Unsaturated nitrile-conjugated diene copolymer process for producing same and ... | Not disclosed | US | 1998 | <https://patents.google.com/patent/US5807941A/en>
- US12359010B2 | Polymer latex for dip-molding applications | Not disclosed | US | 2025 | <https://patents.google.com/patent/US12359010B2/en>
- US4239863A | Process for the polymerization of acrylonitrile-butadiene-styrene resins | Not disclosed | US | 1980 | <https://patents.google.com/patent/US4239863A/en>
- US3903200A | Continuous mass polymerization process for ABS polymeric polyblends | Not disclosed | US | 1975 | <https://patents.google.com/patent/US3903200A/en>
- US11130832B2 | Modified conjugated diene-based polymer and rubber composition including the ... | Not disclosed | US | 2021 | <https://patents.google.com/patent/US11130832B2/en>
- US10336853B2 | Polymer, process and composition | Not disclosed | US | 2019 | <https://patents.google.com/patent/US10336853B2/en>
- US10731020B2 | Rubber composition for vulcanization molding, process for manufacturing the ... | Not disclosed | US | 2020 | <https://patents.google.com/patent/US10731020B2/en>